

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

# AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

NFX301-R

# Netflix's IPv6 journey: Building network configurations on AWS

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AWS



# Agenda

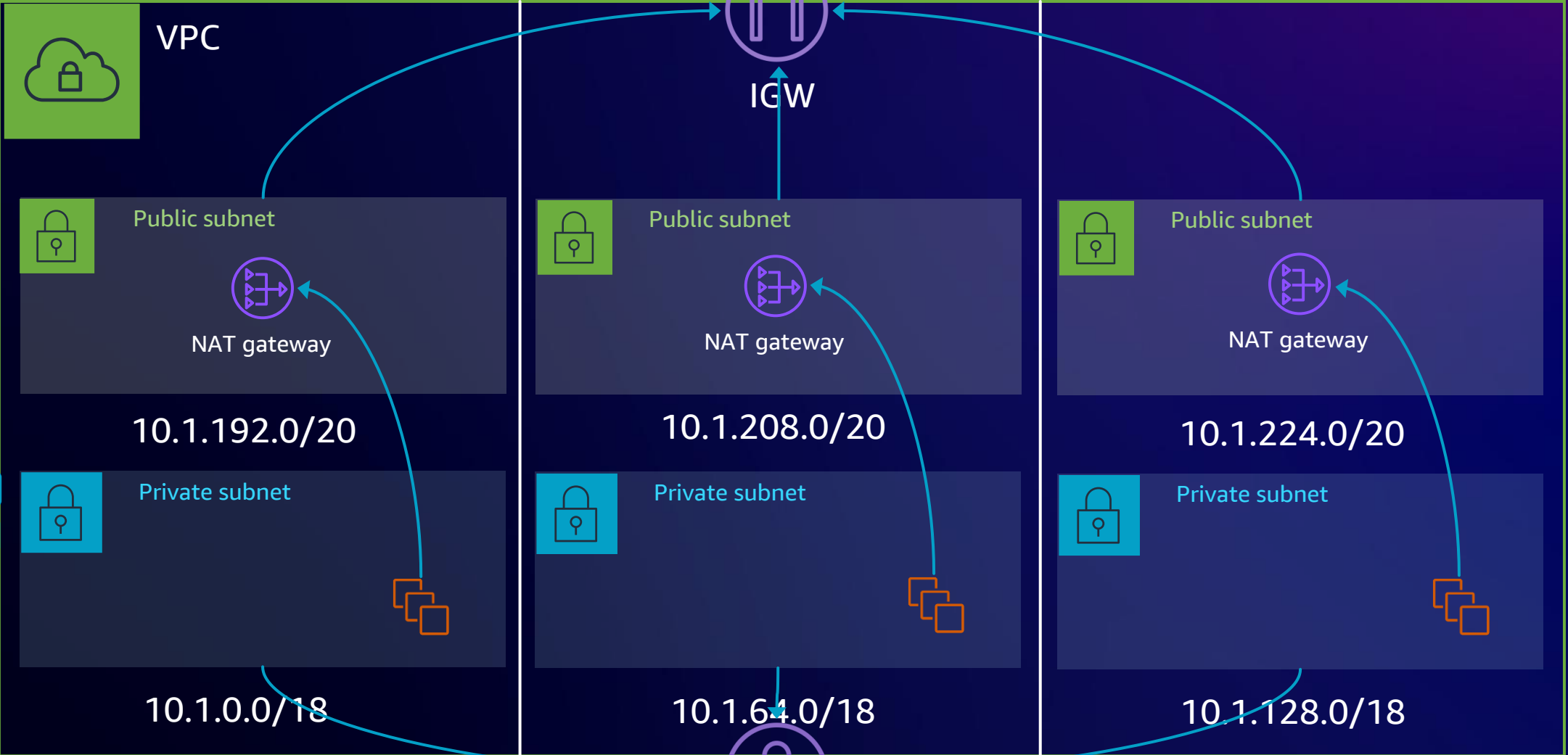
- Why not IPv4?
- Migrating to an IPv6-centric architecture
- What this looked like at Netflix

# Why not IPv4?



AWS Cloud

10.1.0.0/16

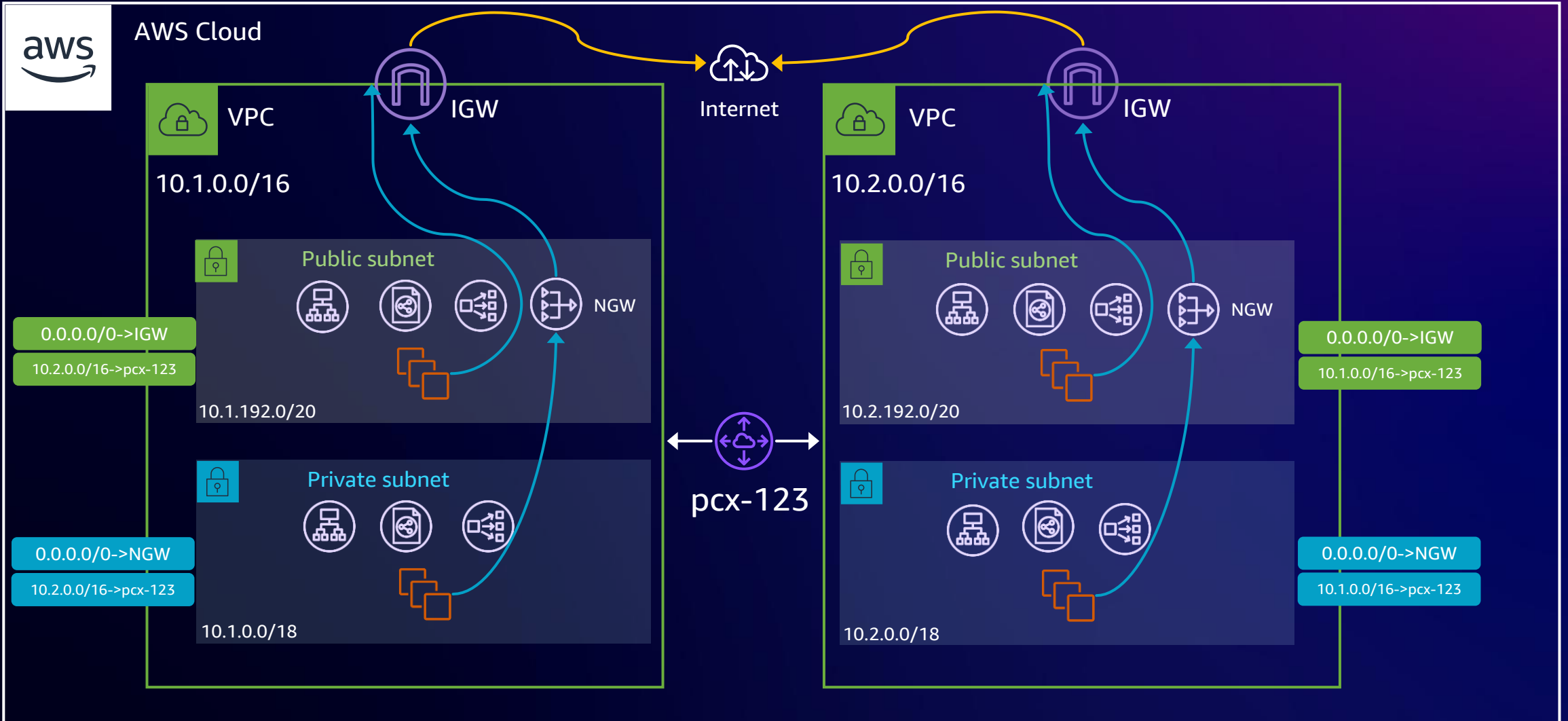


0.0.0.0/0->IGW

0.0.0.0/0->NGW

100.64.0.0/10->VGW

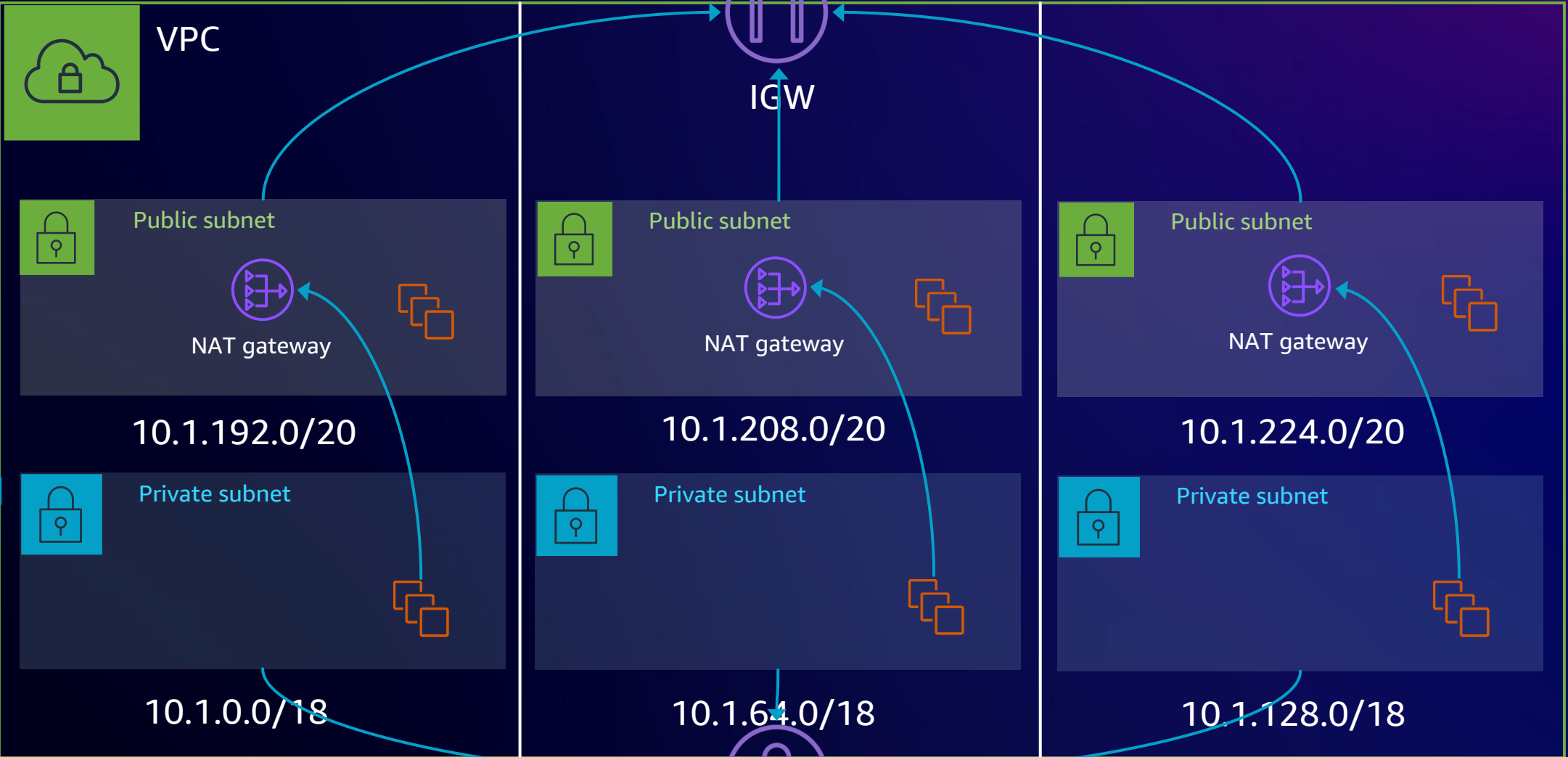






AWS Cloud

10.1.0.0/16



0.0.0.0/0->IGW

::/0->IGW

0.0.0.0/0->NGW

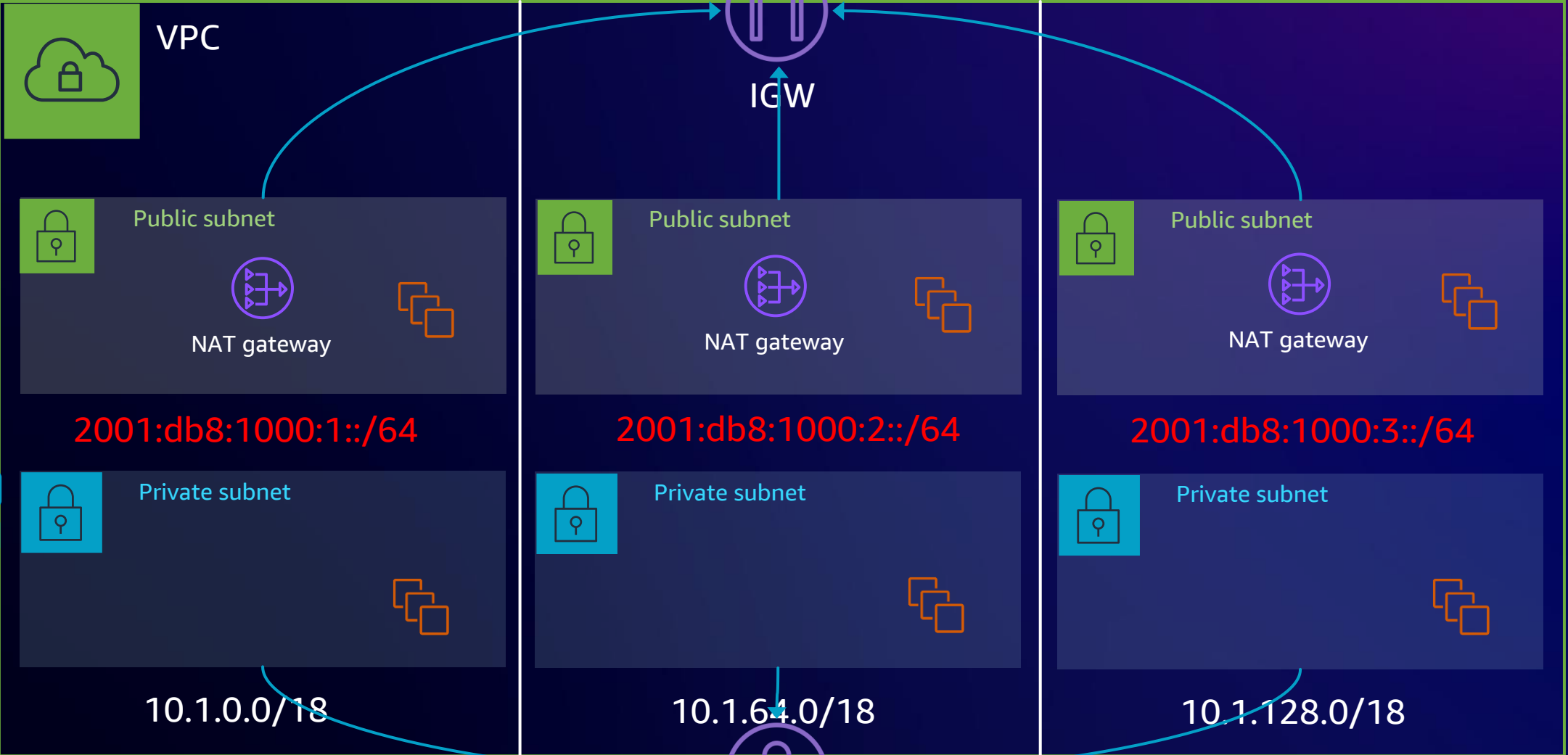
100.64.0.0/10->VGW





AWS Cloud

2001:db8:1000::/56

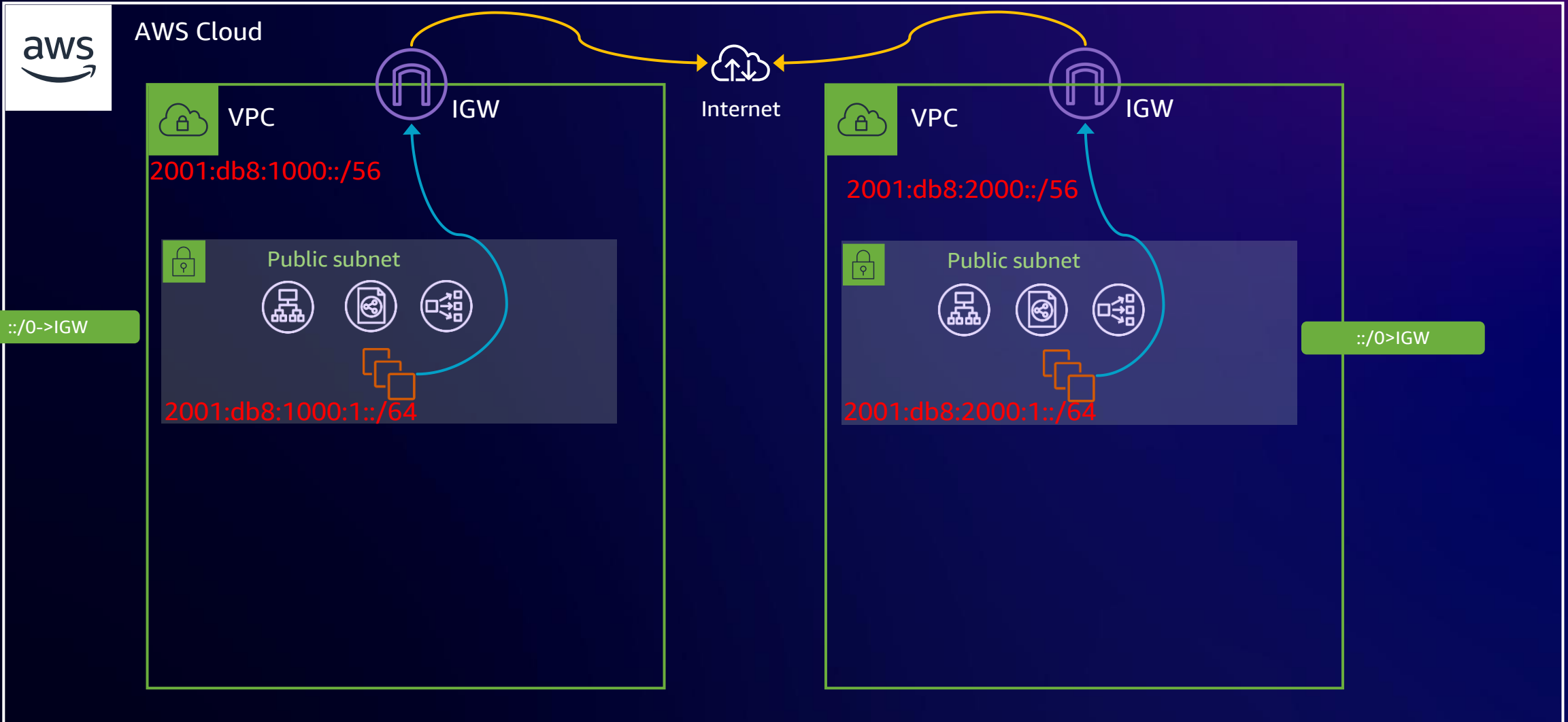


:/0->IGW

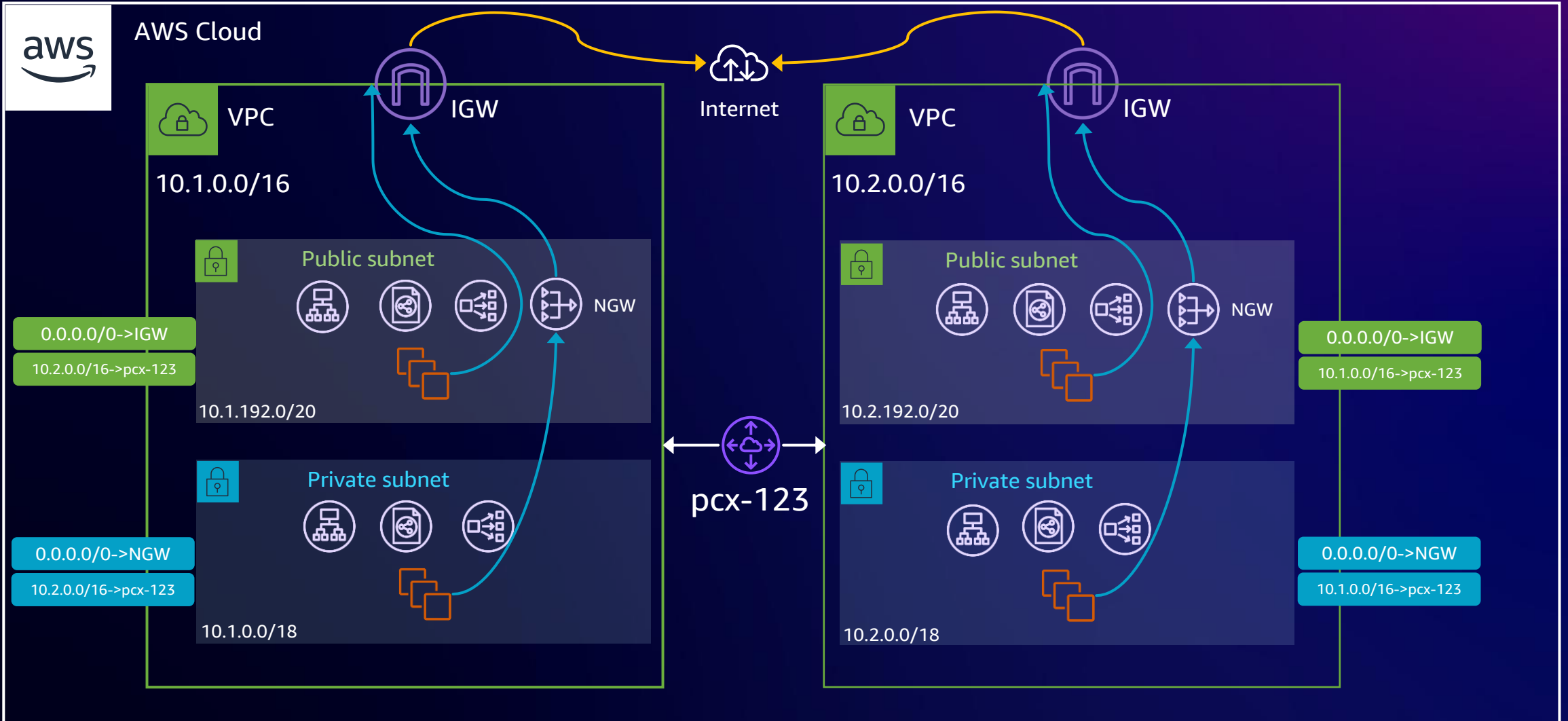
0.0.0.0/0->NGW

100.64.0.0/10->VGW





# Migrating to an IPv6-centric architecture

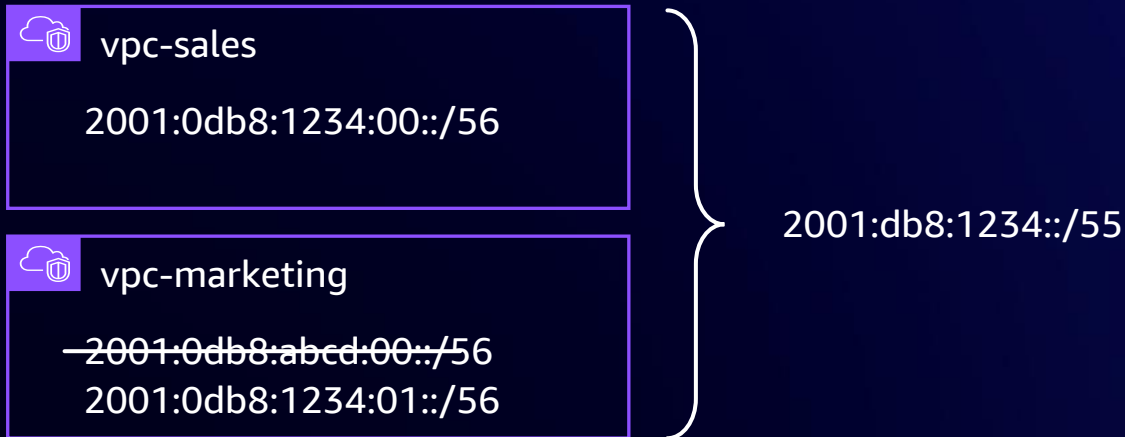


# Dual-stack VPCs

IPv4 only

Dual-stack  
VPC

- AWS provided IPv6 CIDR blocks
  - Provisioned by AWS instantly
  - Not summarizable



### Add IPv6 CIDR

IPv6 CIDR block

☐ IPAM-allocated IPv6 CIDR block

☒ Amazon-provided IPv6 CIDR block

☐ IPv6 CIDR owned by me

Network border group

A network border group is a unique group of Zones from where IPv4 and IPv6 IP addresses are advertised. All Availability Zones in this VPC will use this network border group.

us-east-1

Cancel Select CIDR

# Dual-stack VPCs

IPv4 only

Dual-stack  
VPC

- Bring your own IP (BYOIP)
  - Need custom provision in each account
  - Summarizable

**Add IPv6 CIDR** ×

IPv6 CIDR block

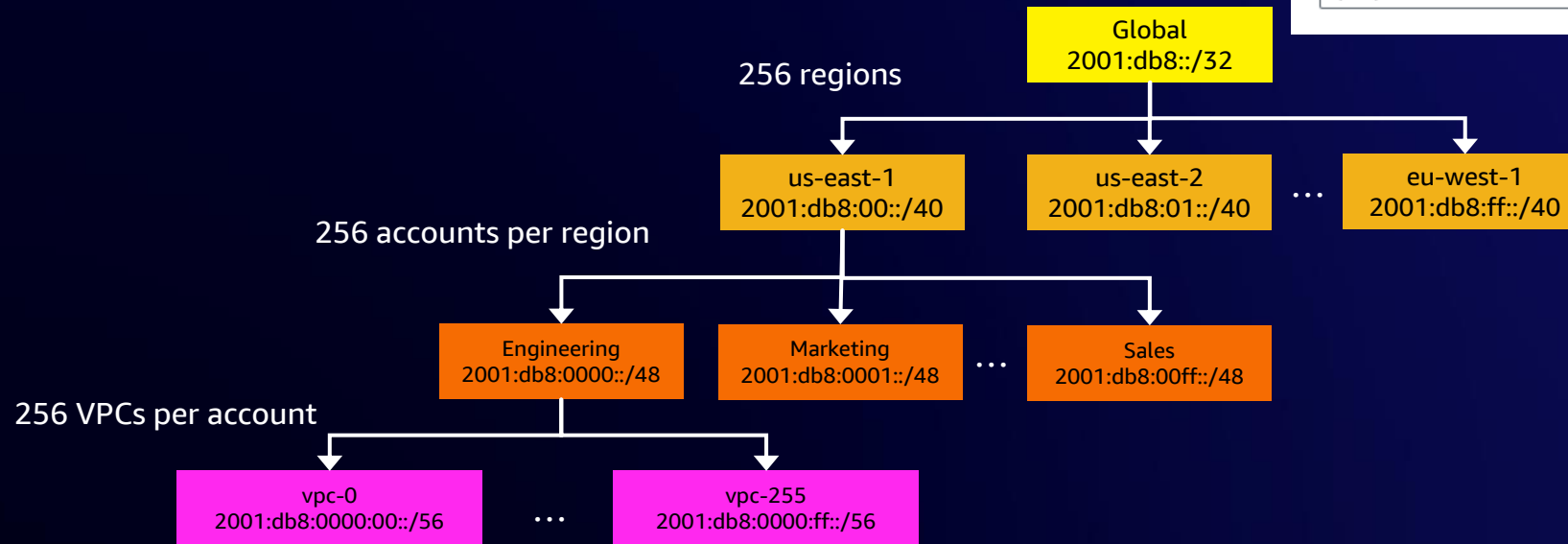
☐ IPAM-allocated IPv6 CIDR block

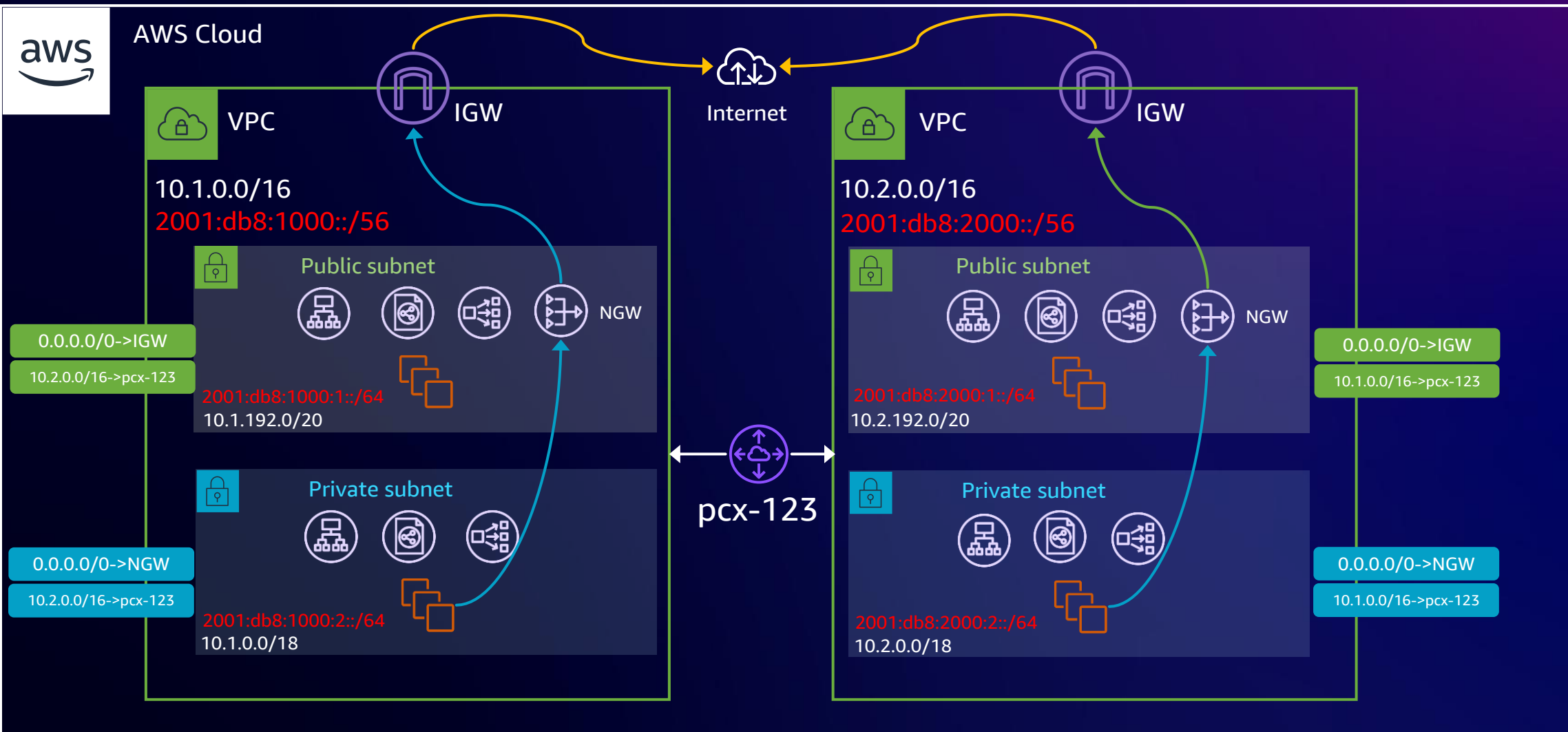
☐ Amazon-provided IPv6 CIDR block

☒ IPv6 CIDR owned by me

Pool

ipv6pool-ec2-02bb0f38a55fce389 ▼





# IPv6-aware network – Security config



Inbound rules

Info

| Type       | Protocol | Port range | Source | Description - optional                                             |                               |
|------------|----------|------------|--------|--------------------------------------------------------------------|-------------------------------|
| Info       | Info     | Info       | Info   | Info                                                               |                               |
| Custom TCP | TCP      | 22         | Custom | <div><div>Q</div><div>sg-045dda3bce7a21ab2</div><div>X</div></div> | <div></div> <div>Delete</div> |
| Custom TCP | TCP      | 22         | Custom | <div><div>Q</div><div>10.1.0.0/16</div><div>X</div></div>          | <div></div> <div>Delete</div> |

# IPv6-aware network – Security config



Inbound rules [Info](#)

| Type <a href="#">Info</a> | Protocol <a href="#">Info</a> | Port range <a href="#">Info</a> | Source <a href="#">Info</a> | Description - optional <a href="#">Info</a>              |                               |
|---------------------------|-------------------------------|---------------------------------|-----------------------------|----------------------------------------------------------|-------------------------------|
| Custom TCP ▼              | TCP                           | 22                              | Custom ▼                    | <div><div>Q</div><div>sg-045dda3bce7a21ab2 X</div></div> | <div></div> <div>Delete</div> |
| Custom TCP ▼              | TCP                           | 22                              | Custom ▼                    | <div><div>Q</div><div>10.1.0.0/16 X</div></div>          | <div></div> <div>Delete</div> |
| Custom TCP ▼              | TCP                           | 22                              | Custom ▼                    | <div><div>Q</div><div>2001:db8:1000::/56 X</div></div>   | <div></div> <div>Delete</div> |

Add rule

# IPv6-aware network – Routing config

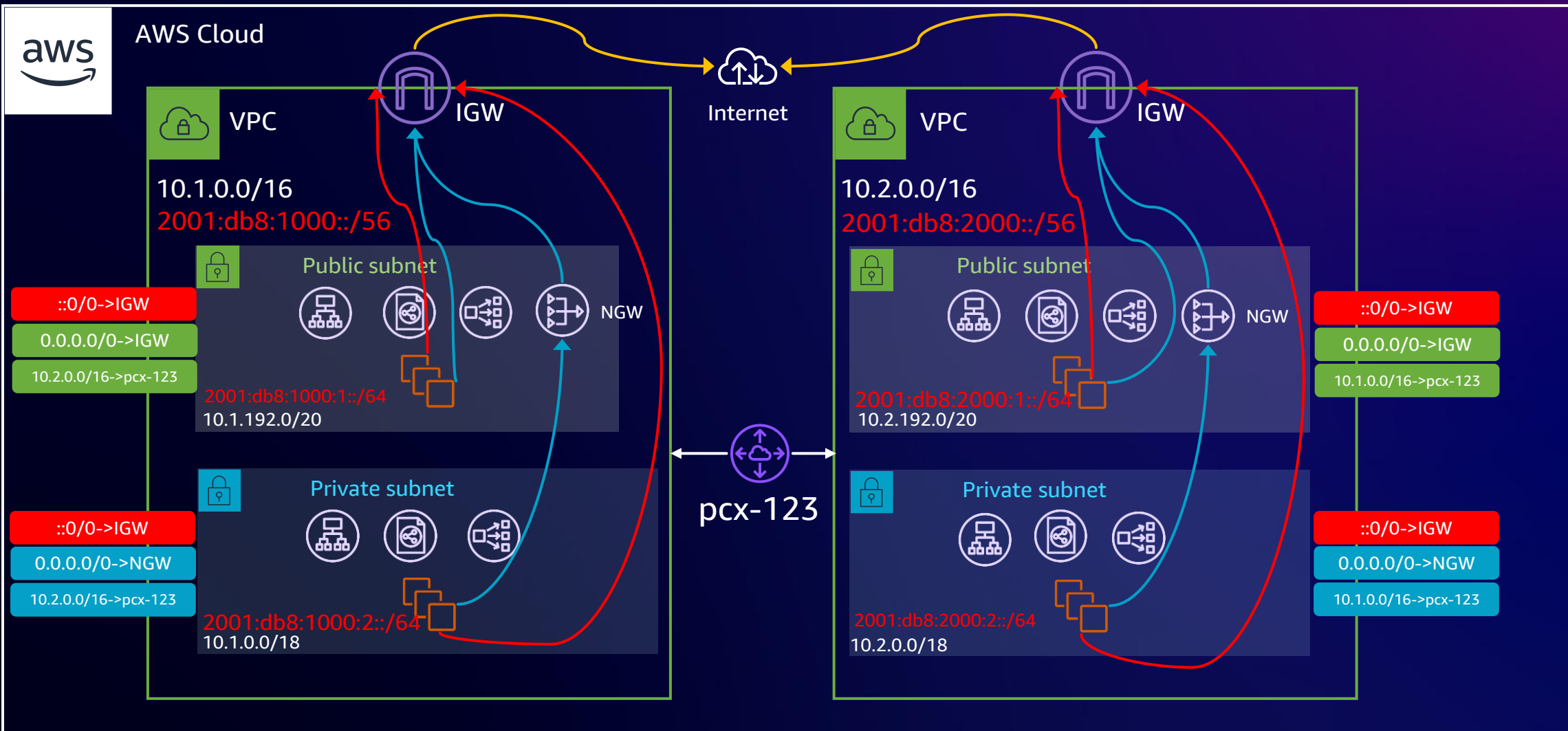


# IPv6-aware network – Routing config



|                                            |                                               |   |
|--------------------------------------------|-----------------------------------------------|---|
| <input type="text" value="0.0.0.0/0"/>     | <input type="text" value="Internet Gateway"/> | - |
| <input type="text" value="igw-6dfdca08"/>  |                                               |   |
| <input "::="" 0"="" type="text" value=""/> | <input type="text" value="Internet Gateway"/> | - |
| <input type="text" value="igw-6dfdca08"/>  |                                               |   |
| <input type="button" value="Add route"/>   |                                               |   |



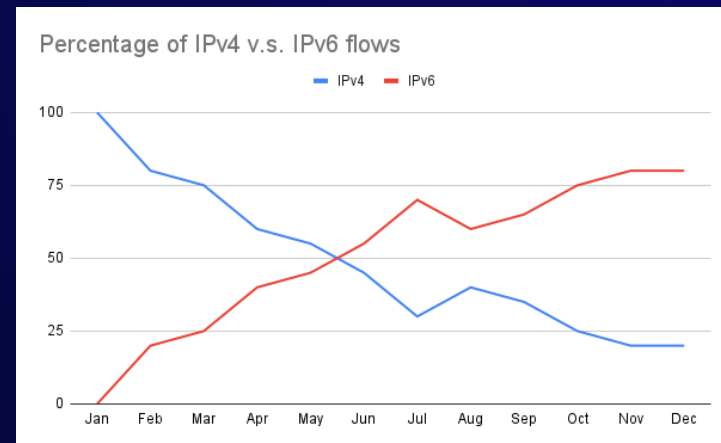
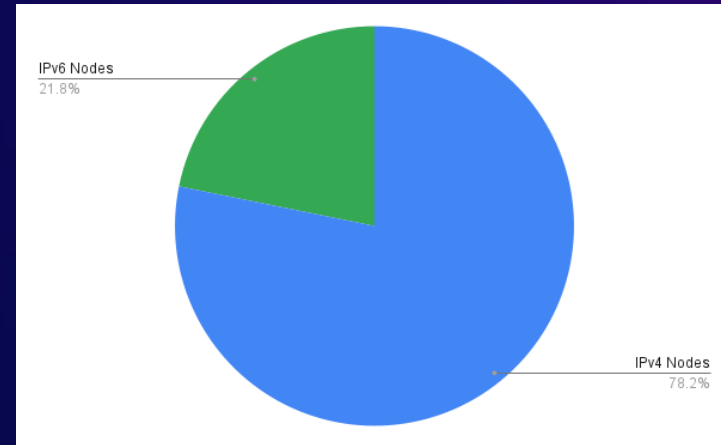


→ IPv6 route  
→ IPv4 route

# Enable IPv6 – Observability



- Track and measure your progress
- Define metrics and build dashboard
  - % of nodes enabled with IPv6 vs. IPv4
  - % of flows using IPv6 vs. IPv4
  - Group by app



# Enable IPv6 – Identify apps to start



- Define migration strategy
- Smaller apps are safer
  - Gain confidence before changing business-critical apps
- Bigger apps yield more signals
  - More network usage
  - More likely to find and fix issues early

# Enable IPv6 – Communication



- Notify app owners before any change
- Avoid surprise

# Enable IPv6 – Amazon EC2

IPv4 only

Dual-stack  
VPC

IPv6-aware  
network

Enable  
IPv6

## AWS::EC2::LaunchTemplate NetworkInterface

RSS

Specifies the parameters for a network interface.

NetworkInterface is a property of `AWS::EC2::LaunchTemplate` `LaunchTemplateData`.

### Syntax

To declare this entity in your AWS CloudFormation template, use the following syntax:

### JSON

```
{
  "AssociateCarrierIpAddress" : Boolean,
  "AssociatePublicIpAddress" : Boolean,
  "DeleteOnTermination" : Boolean,
  "Description" : String,
  "DeviceIndex" : Integer,
  "Groups" : [ String, ... ],
  "InterfaceType" : String,
  "Ipv4PrefixCount" : Integer,
  "Ipv4Prefixes" : [ Ipv4PrefixSpecification, ... ],
  "Ipv6AddressCount" : Integer,
  "Ipv6Addresses" : [ Ipv6Add, ... ],
  "Ipv6PrefixCount" : Integer,
  "Ipv6Prefixes" : [ Ipv6PrefixSpecification, ... ],
  "NetworkCardIndex" : Integer,
  "NetworkInterfaceId" : String,
  "PrimaryIpv6" : Boolean,
  "PrivateIpAddress" : String,
  "PrivateIpAddresses" : [ PrivateIpAdd, ... ],
  "SecondaryPrivateIpAddressCount" : Integer,
  "SubnetId" : String
}
```

**Associate IPv6 (Recommended)  
Address**

☒ Assign an IPv6 address to instances

- Make a new deployment
  - Monitor % of IPv6 nodes metric

# Enable IPv6 – Load balancer



**Basic configuration**

Select the IP address type of your load balancer based on the CIDR blocks of its subnets. A dual-stack IP address requires both IPv4 and IPv6 CIDR blocks for its subnets.

IP address type

Info

Select the type of IP addresses that your subnets use.

☐ IPv4

Recommended for internal load balancers.

☒ Dualstack

Includes IPv4 and IPv6 addresses.

▼ Record 1

Delete

Record name

Info

reinvent.demo

.test.netflix.net

Record type

Info

CNAME – Routes traffic to another domain name and to some AWS reso... ▼

Keep blank to create a record for the root domain.

☒ Alias

▼ Record 1

Delete

Record name

Info

reinvent.demo

.test.netflix.net

Record type

Info

AAAA – Routes traffic to an IPv6 address and some AWS resources ▼

Keep blank to create a record for the root domain.

☒ Alias



# Enable IPv6 – Service discovery



# Use IPv6



- IPv6 enabled != IPv6 used
- Bind to IPv6 address
  - -Djava.net.PreferIPv4Stack=false
- Make sure DNS resolves to IPv6
  - Load balancer DNS AAAA aliases
  - S3 DNS rewritten by CoreDNS

# Use IPv6

IPv4 only

Dual-stack  
VPC

IPv6-aware  
network

Enable  
IPv6

Use IPv6

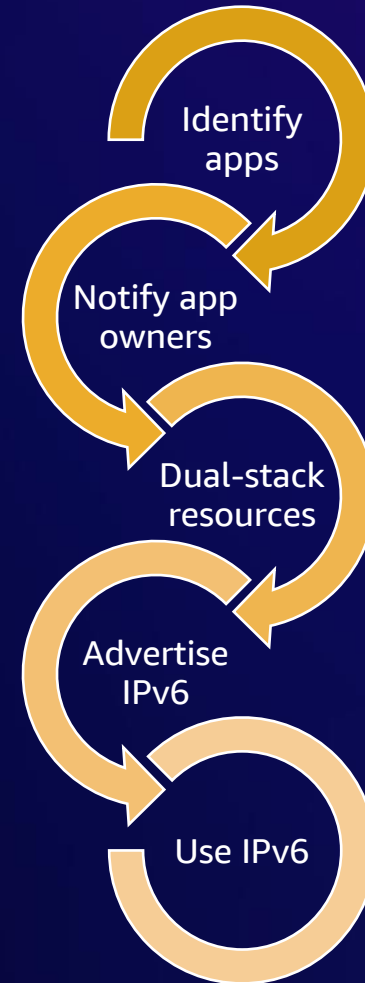
```
# it appears IPv6 S3 is supported in all regions, so we can use a simple regex
rewrite continue {
    name regex s3\.(.*)\.amazonaws\.com s3.dualstack.{1}.amazonaws.com
    answer name s3\.dualstack\.(.*)\.amazonaws\.com s3.{1}.amazonaws.com
}

# for virtual hosting of buckets in a region (i.e. <bucket>.s3.<region>.amazonaws.com.)
rewrite continue {
    name regex (.*)\.s3\.(.*)\.amazonaws\.com {1}.s3.dualstack.{2}.amazonaws.com
    answer name (.*)\.s3\.dualstack\.(.*)\.amazonaws\.com {1}.s3.{2}.amazonaws.com
}
```

# Use IPv6 – Workflow engine



- Automate and streamline the process

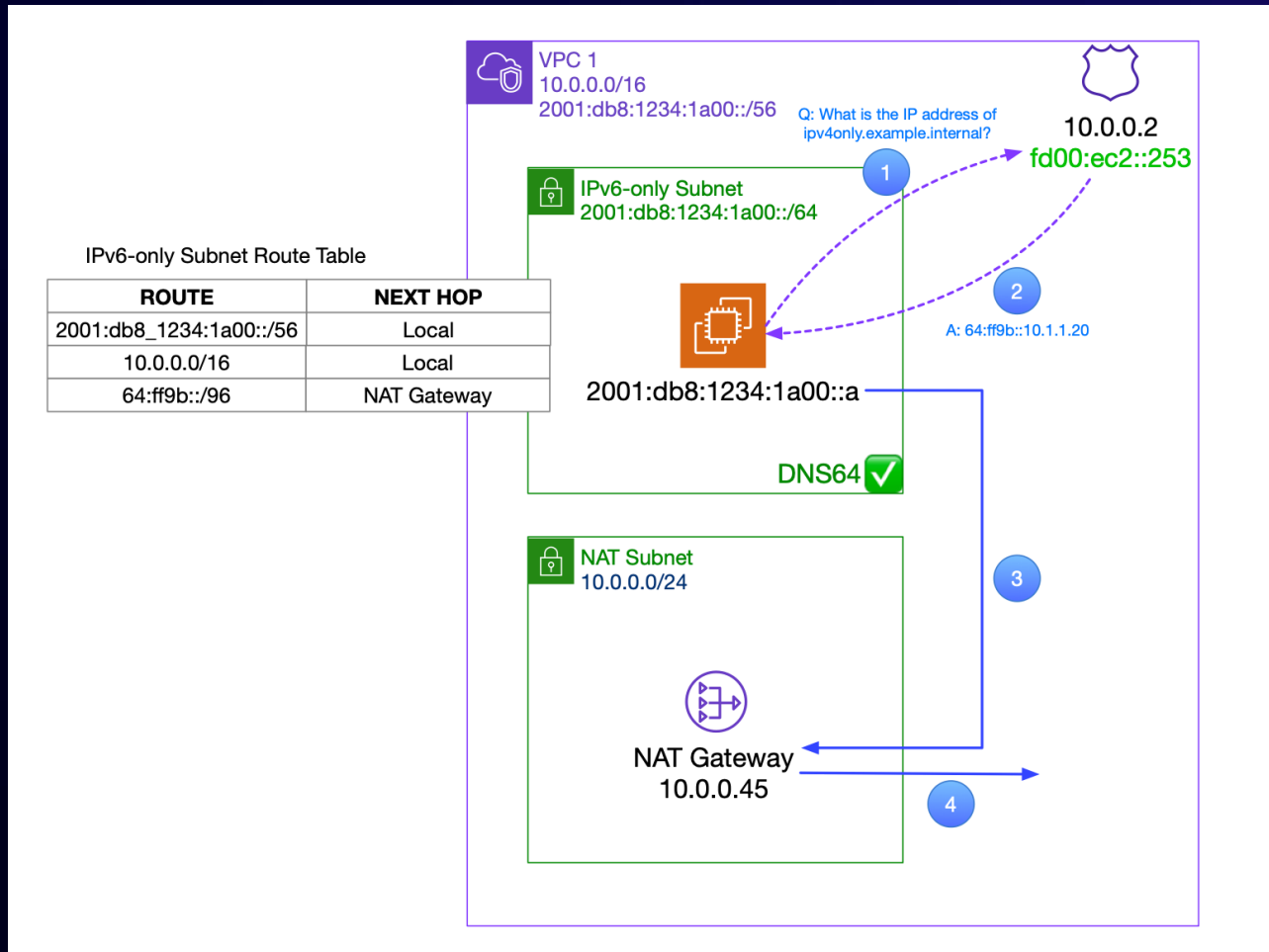


# IPv6 only – Transition mechanism



- IPv4 external dependencies
  - Third-party dependencies
  - AWS services not supporting IPv6

# IPv6 only – Transition mechanism DNS64/NAT64



# IPv6 only – Transition mechanism TSA

IPv4 only

Dual-stack  
VPC

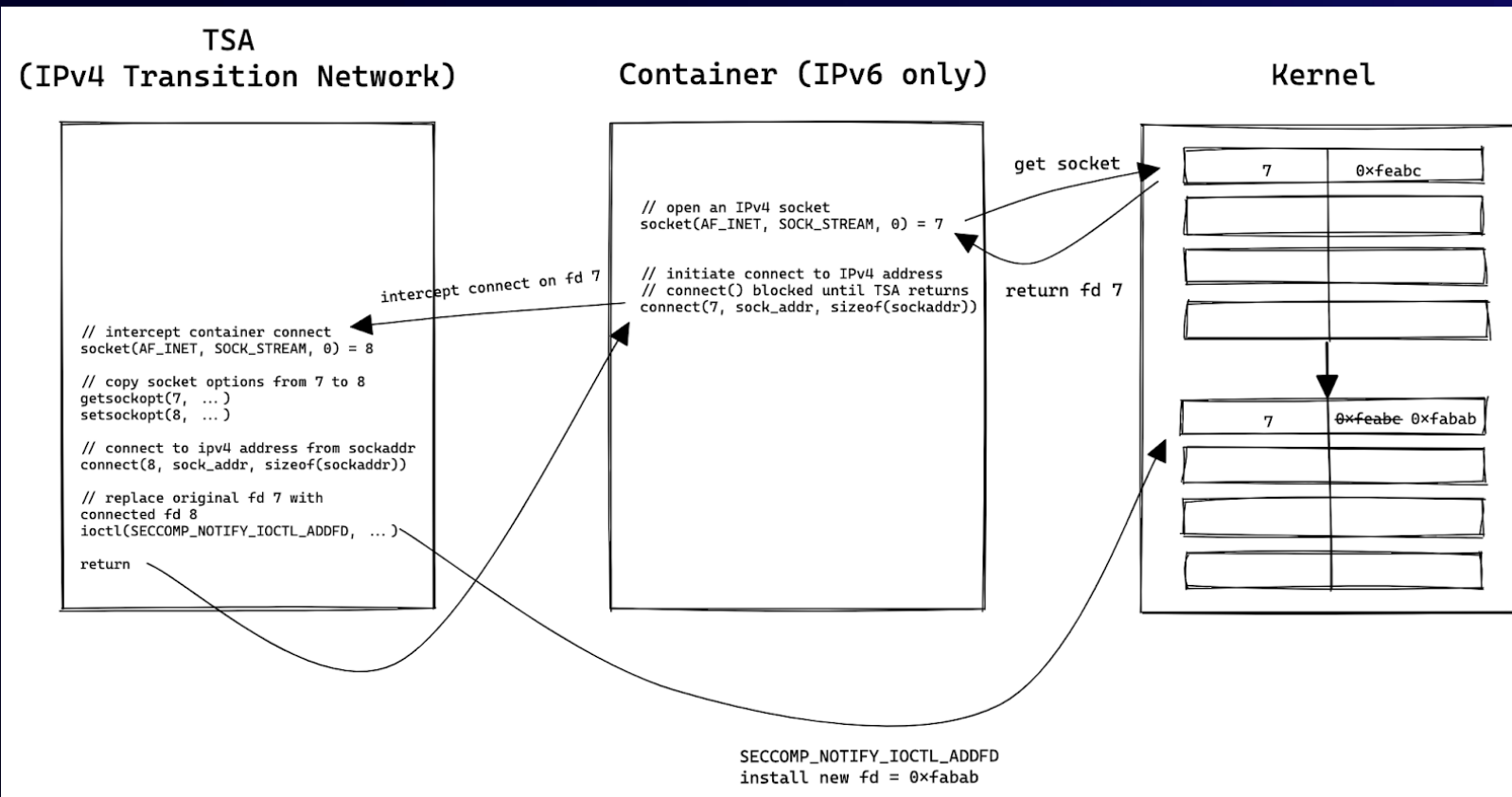
IPv6-aware  
network

Enable  
IPv6

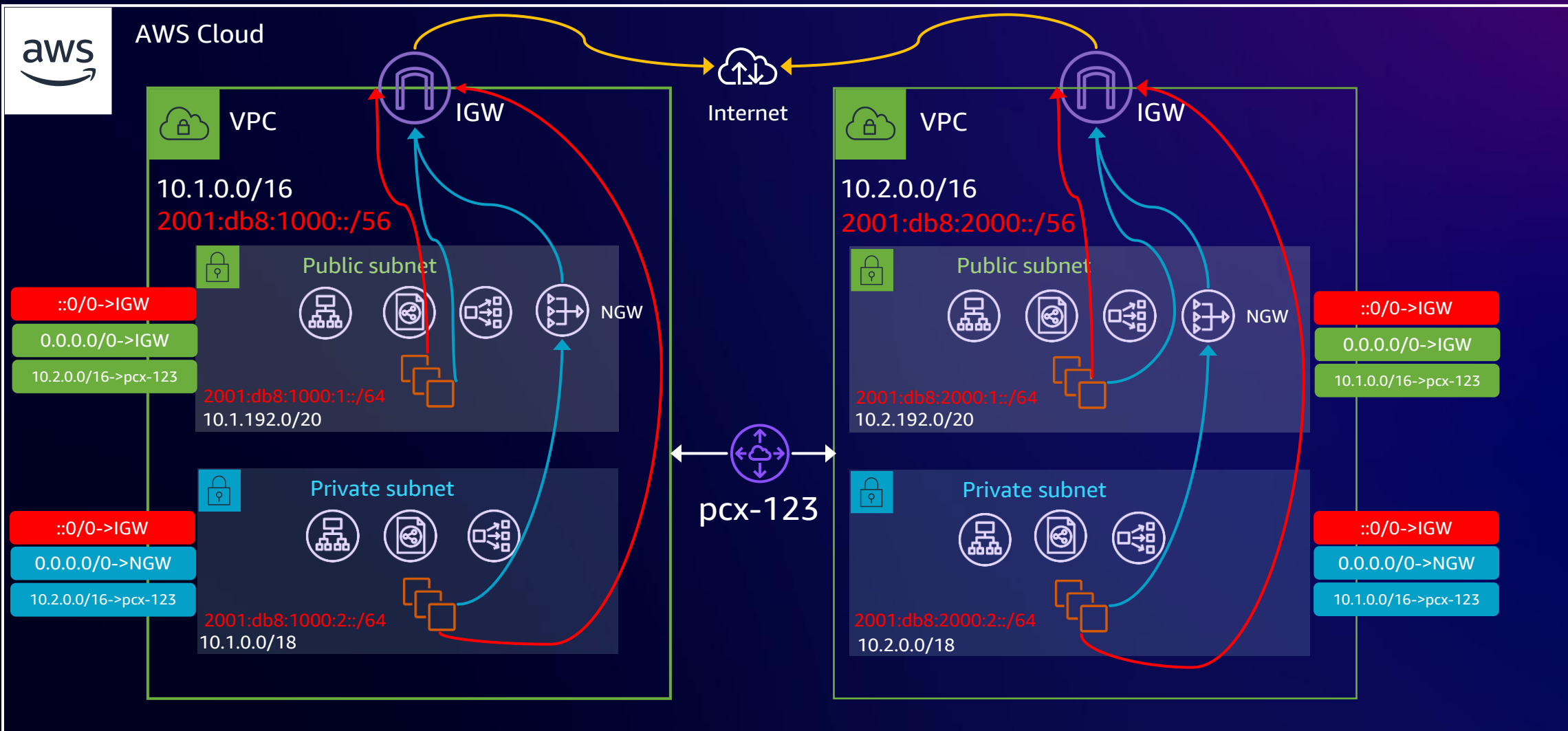
Use IPv6

IPv6 only

- Titus Seccomp Agent



TSA presentation  
<https://bit.ly/nflx-tsa>



— IPv6 route  
— IPv4 route

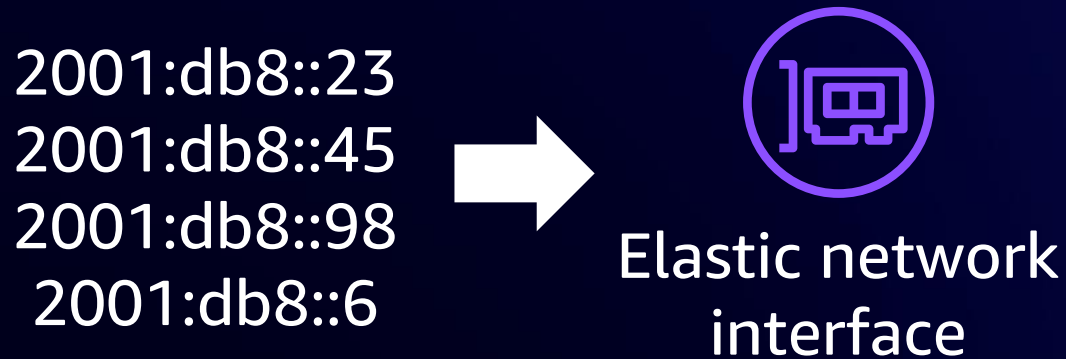
# What this looked like at Netflix

# Netflix IPv6 journey

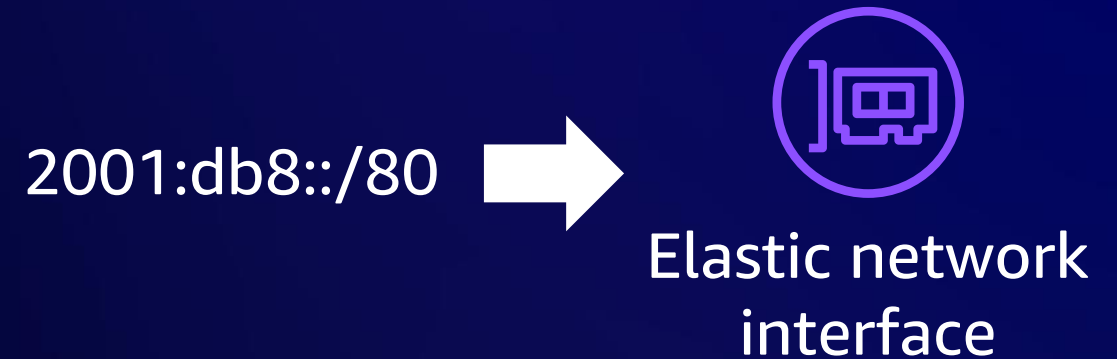
1. Strategy, communication, and planning
2. Execution: Scale, cost savings

# Netflix IPv6 journey – Strategy

- Scale the VPC network
  1. Network address usage (NAU)
  2. Private IPv4 address allocations
  3. “Dynamism” – How fast we can change the VPC network

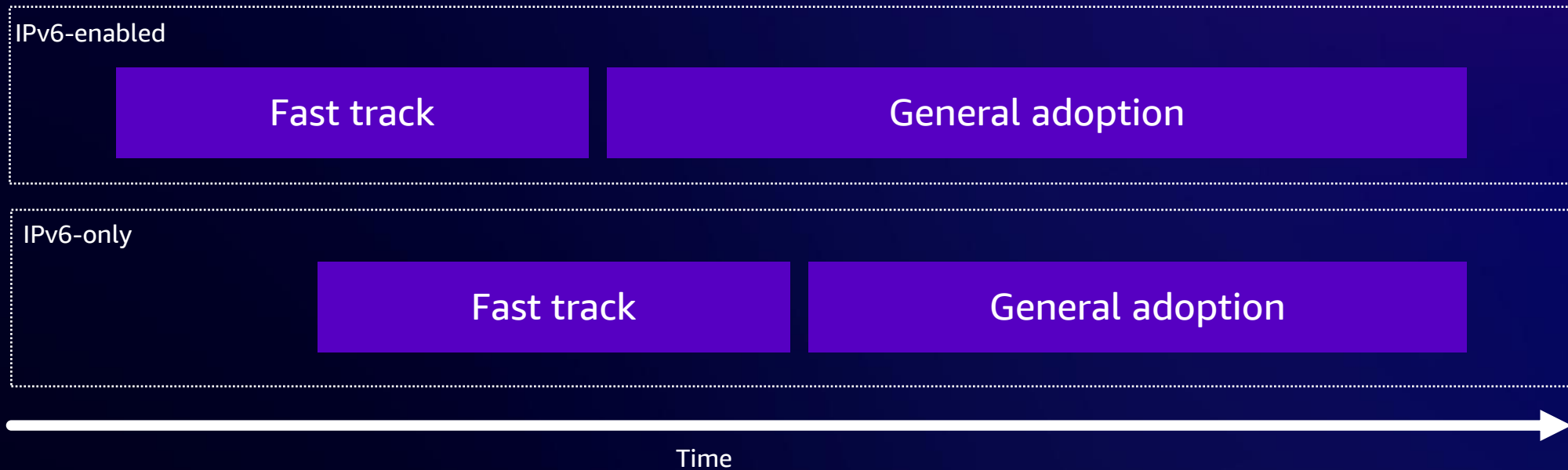


Individual IP assignments



Prefix delegation

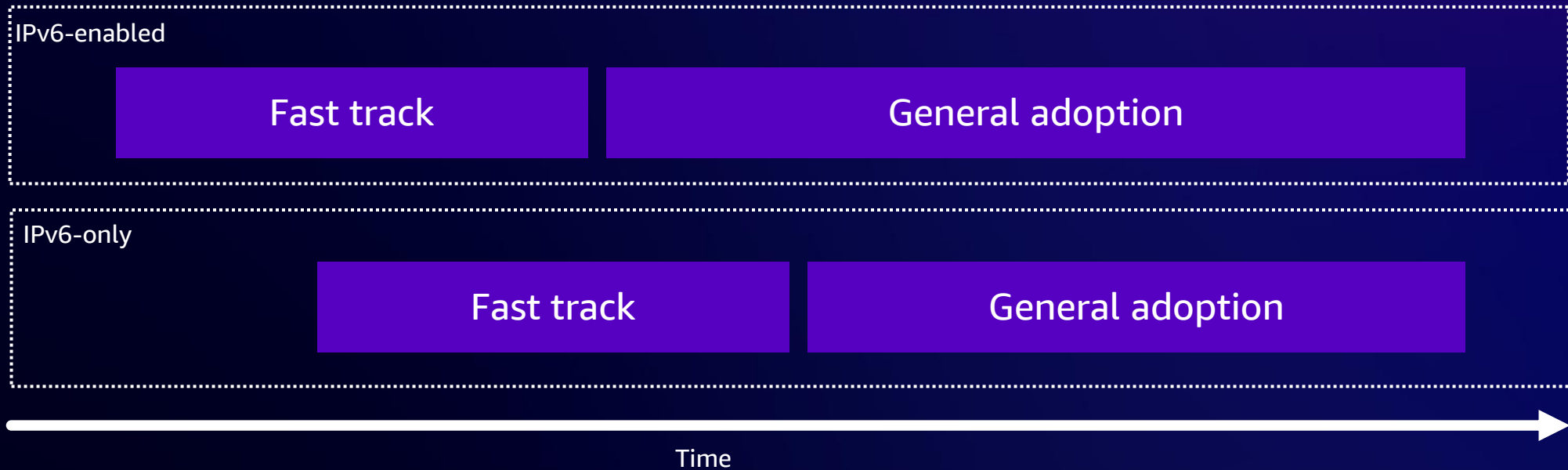
# Netflix IPv6 journey – Planning



# Netflix IPv6 journey – Communication



# Netflix IPv6 journey – Execution: Scale

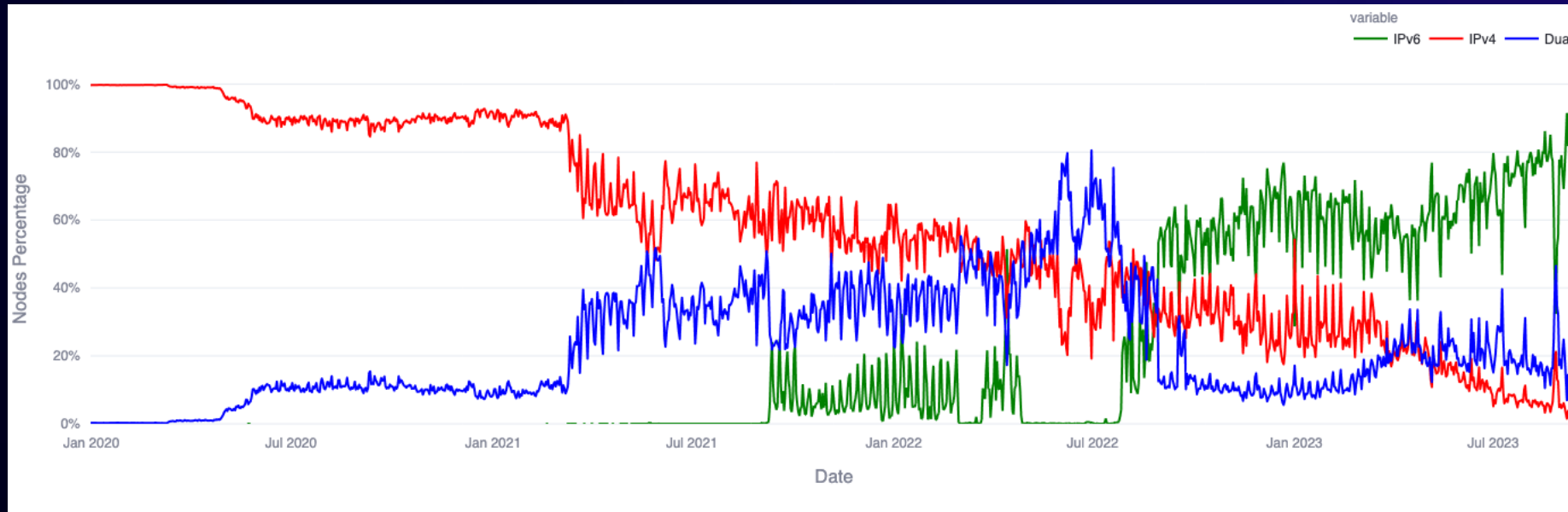


# Netflix IPv6 journey – Execution: Scale

IPv6-enabled

Fast track

General adoption

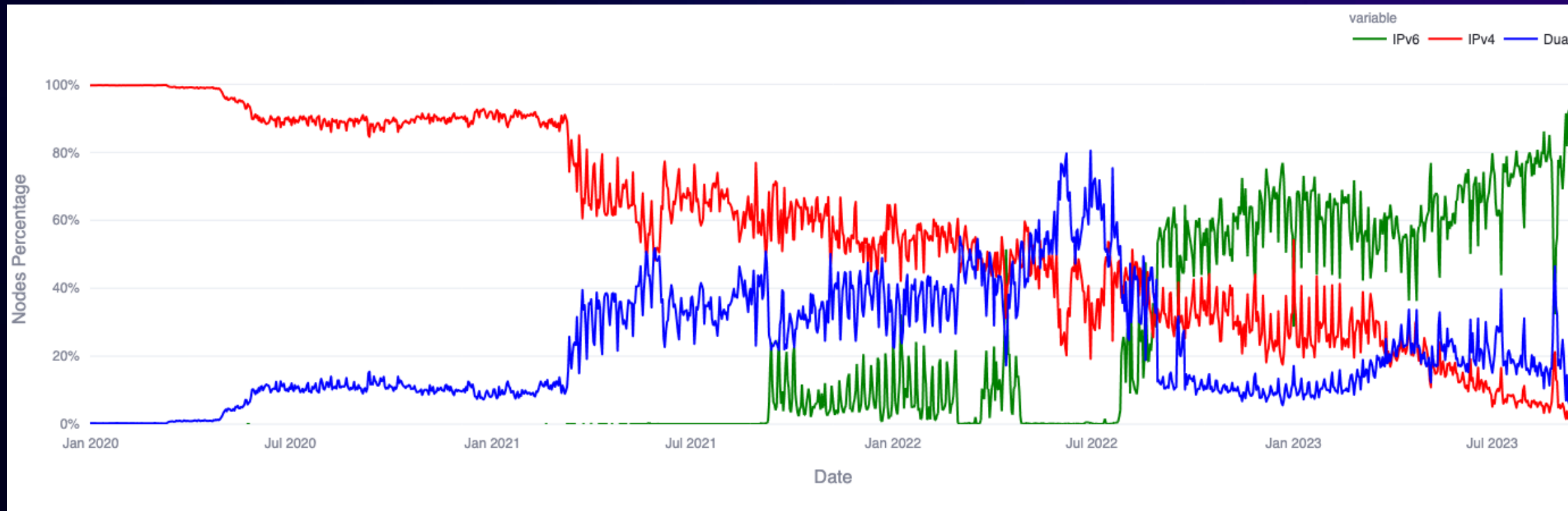


IPv6-only

Fast track

General adoption

# Netflix IPv6 journey – Execution: Scale



# Netflix IPv6 journey – Execution: Scale

IPv6-only\*

~93%

IPv6-enabled\*

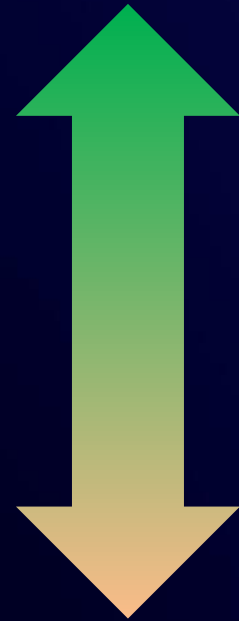
~99%

IPv4-only\*

~1%

# Netflix IPv6 journey – Execution: Cost savings

The cost of moving bytes\*



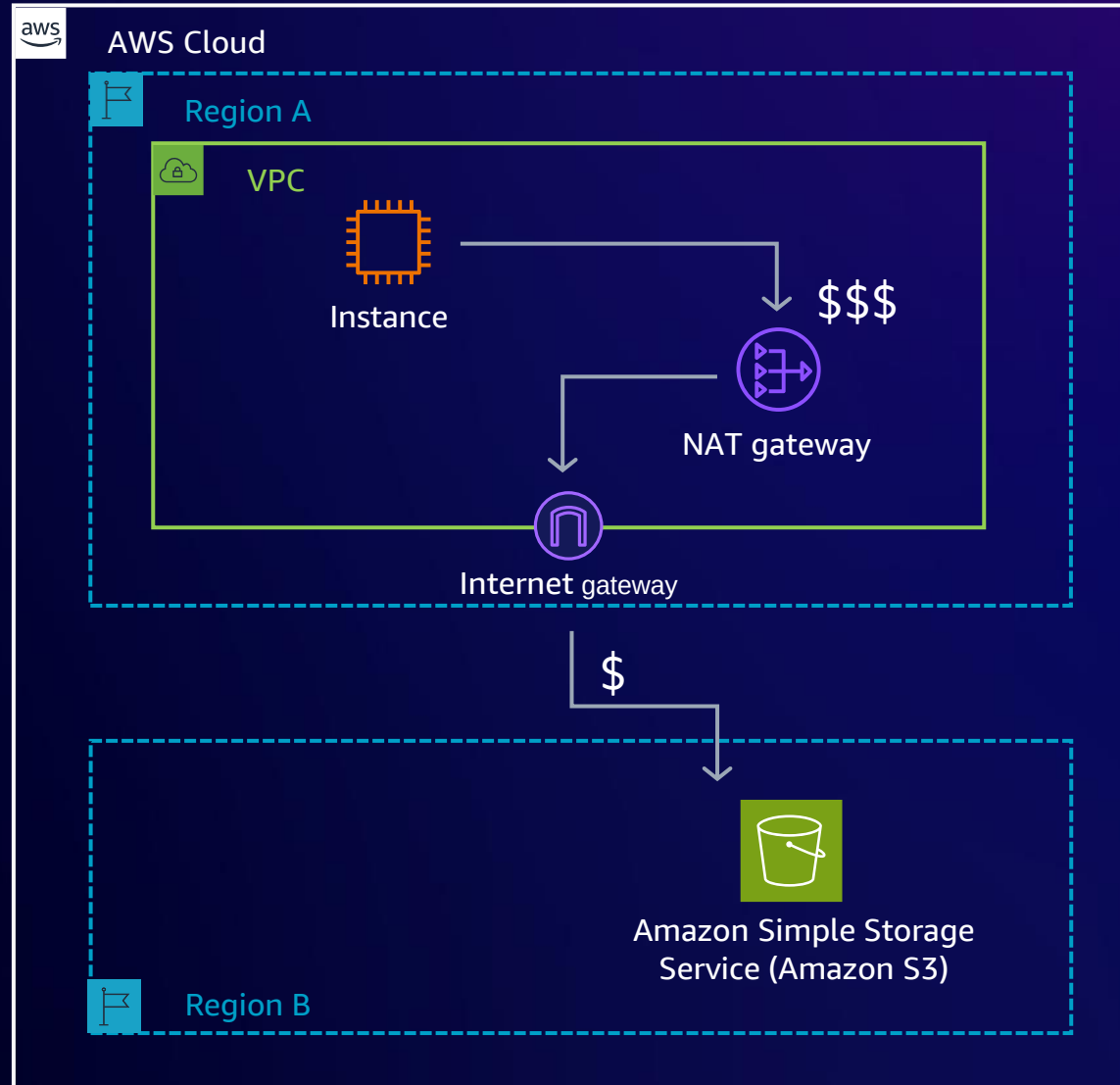
Amazon Virtual Private Cloud (Amazon VPC) peering  
Internet gateway (IGW)  
AWS PrivateLink/Amazon VPC endpoints  
AWS Direct Connect  
Transit gateway (TGW)  
NAT Gateway (NGW)

# Netflix IPv6 journey – Execution: Cost savings

## Cross-Region S3 access

IPv4 (IPv4 Public access, NAT)

|          |                            |
|----------|----------------------------|
| \$\$\$   | NAT Gateway                |
| + \$     | Data Transfer Inter-region |
| <hr/>    |                            |
| \$\$\$\$ | Total                      |



# Netflix IPv6 journey – Execution: Cost savings

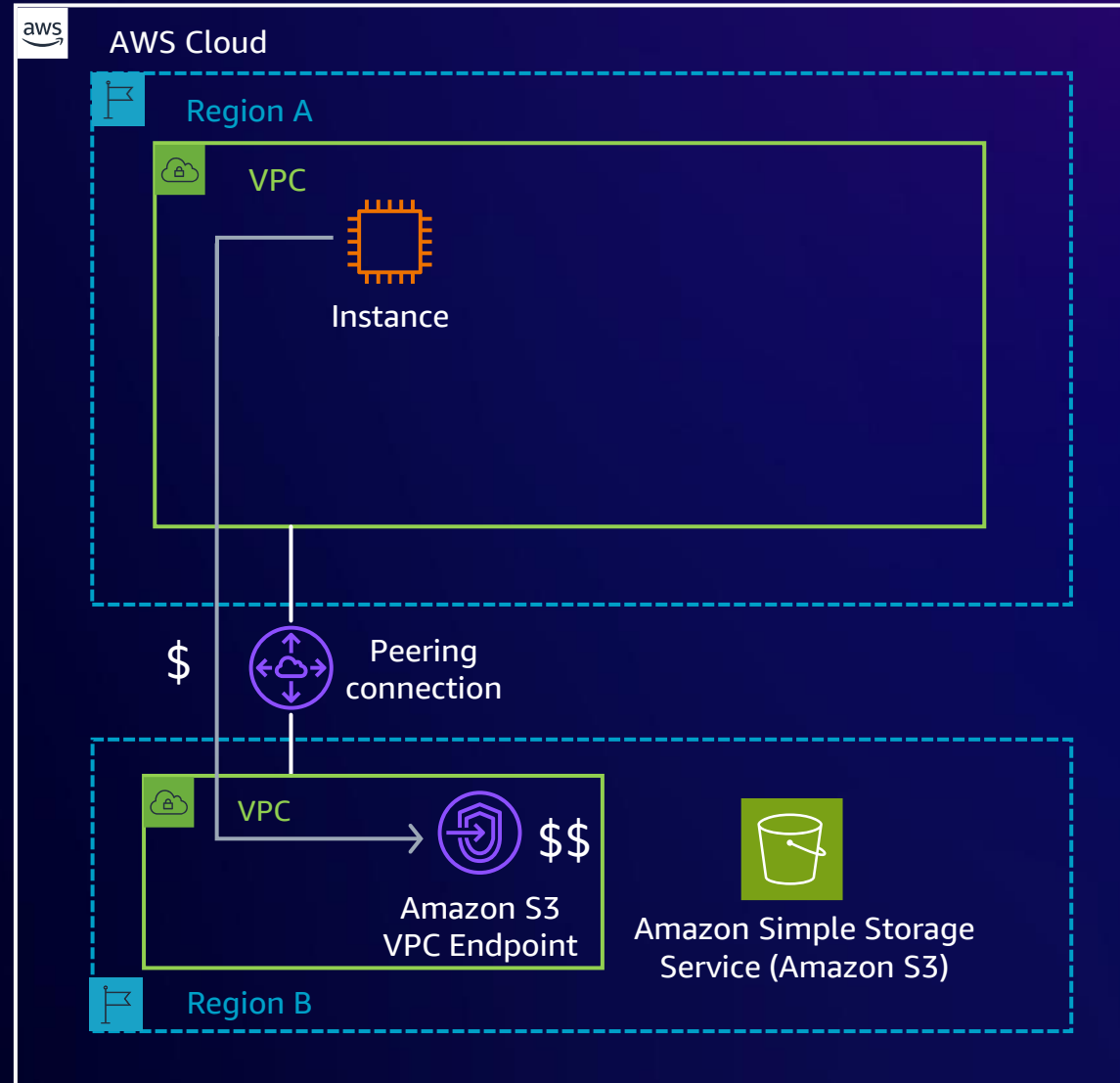
## Cross-Region S3 access

### IPv4 (IPv4 Public access, NAT)

|       |          |                            |
|-------|----------|----------------------------|
|       | \$\$\$   | NAT Gateway                |
| +     | \$       | Data Transfer Inter-region |
| <hr/> |          |                            |
|       | \$\$\$\$ | Total                      |

### IPv4 (IPv4 Private access)

|       |        |                            |
|-------|--------|----------------------------|
|       | \$\$   | VPC Endpoint               |
| +     | \$     | Data Transfer Inter-region |
| <hr/> |        |                            |
|       | \$\$\$ | Total                      |



# Netflix IPv6 journey – Execution: Cost savings

## Cross-Region S3 access

### IPv4 (IPv4 Public access, NAT)

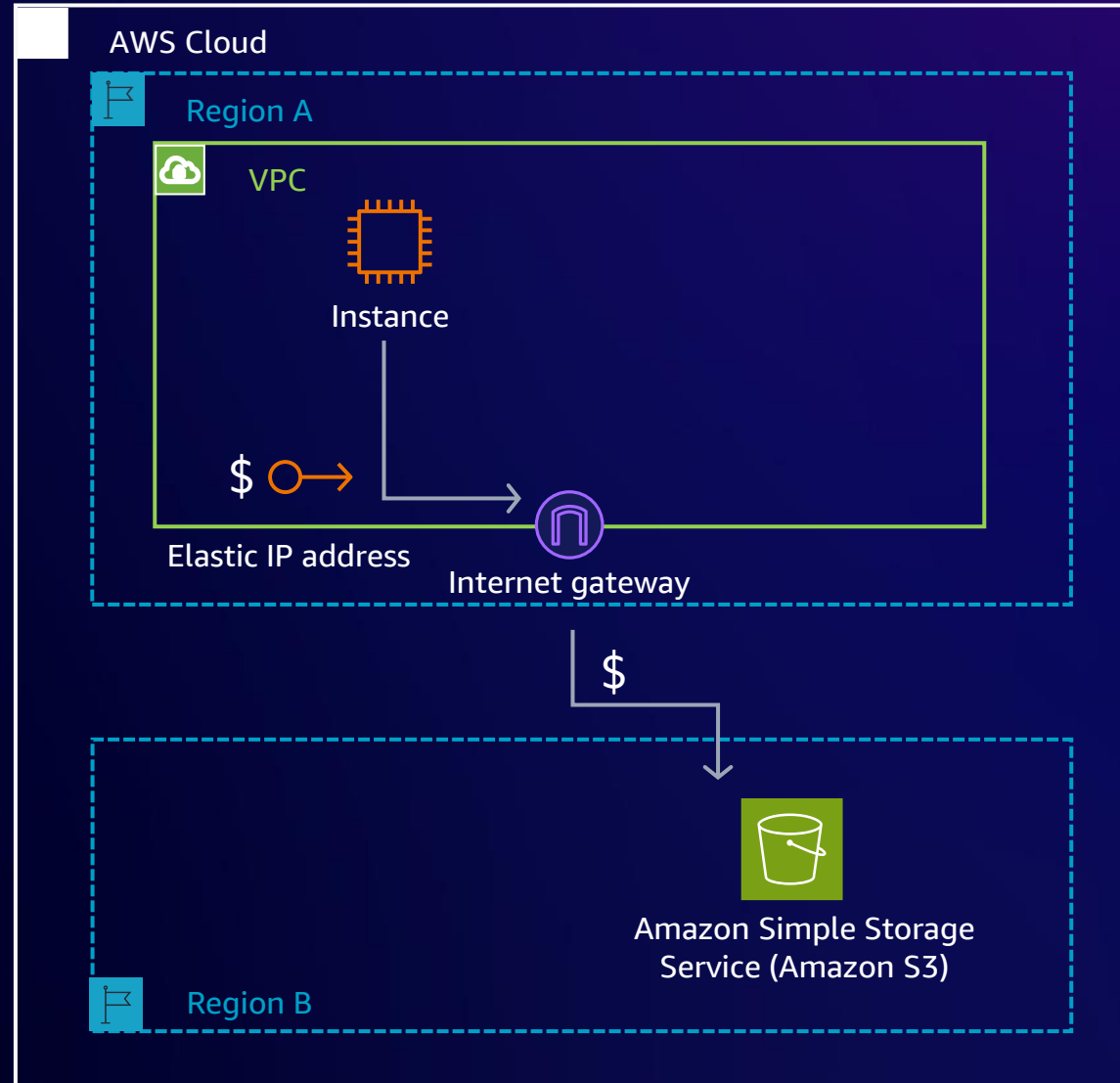
|       |          |                            |
|-------|----------|----------------------------|
|       | \$\$\$   | NAT Gateway                |
| +     | \$       | Data Transfer Inter-region |
| <hr/> |          |                            |
|       | \$\$\$\$ | Total                      |

### IPv4 (IPv4 Private access)

|       |        |                            |
|-------|--------|----------------------------|
|       | \$\$   | VPC Endpoint               |
| +     | \$     | Data Transfer Inter-region |
| <hr/> |        |                            |
|       | \$\$\$ | Total                      |

### IPv4 (IPv4 Public access, EIP)

|       |      |                            |
|-------|------|----------------------------|
|       | \$   | Elastic IP                 |
| +     | \$   | Data Transfer Inter-region |
| <hr/> |      |                            |
|       | \$\$ | Total                      |



# Netflix IPv6 journey – Execution: Cost savings

## Cross-Region S3 access

### IPv4 (IPv4 Public access, NAT)

|       |          |                            |
|-------|----------|----------------------------|
|       | \$\$\$   | NAT Gateway                |
| +     | \$       | Data Transfer Inter-region |
| <hr/> |          |                            |
|       | \$\$\$\$ | Total                      |

### IPv4 (IPv4 Private access)

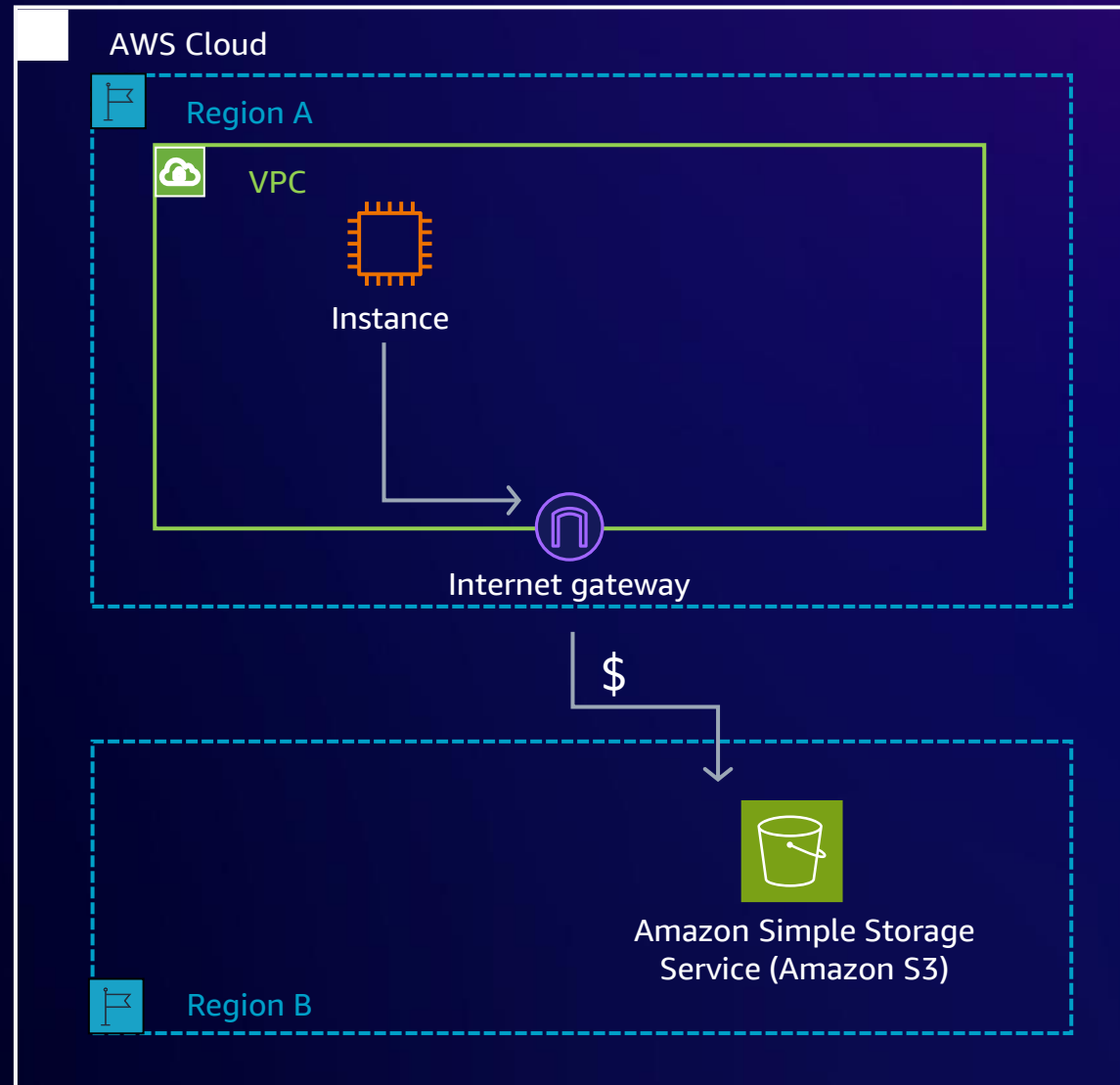
|       |      |                            |
|-------|------|----------------------------|
|       | \$   | VPC Endpoint               |
| +     | \$   | Data Transfer Inter-region |
| <hr/> |      |                            |
|       | \$\$ | Total                      |

### IPv4 (IPv4 Public access, EIP)

|       |      |                            |
|-------|------|----------------------------|
|       | \$   | Elastic IP                 |
| +     | \$   | Data Transfer Inter-region |
| <hr/> |      |                            |
|       | \$\$ | Total                      |

### IPv6

|       |    |                            |
|-------|----|----------------------------|
| +     | \$ | Data Transfer Inter-region |
| <hr/> |    |                            |
|       | \$ | Total                      |



# Thank you!



Please complete the session  
survey in the mobile app